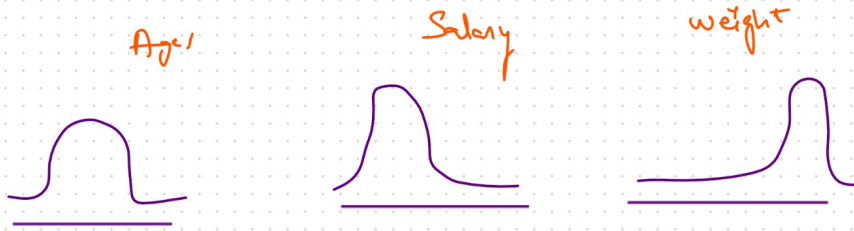


Statistics

Types of probability distribution



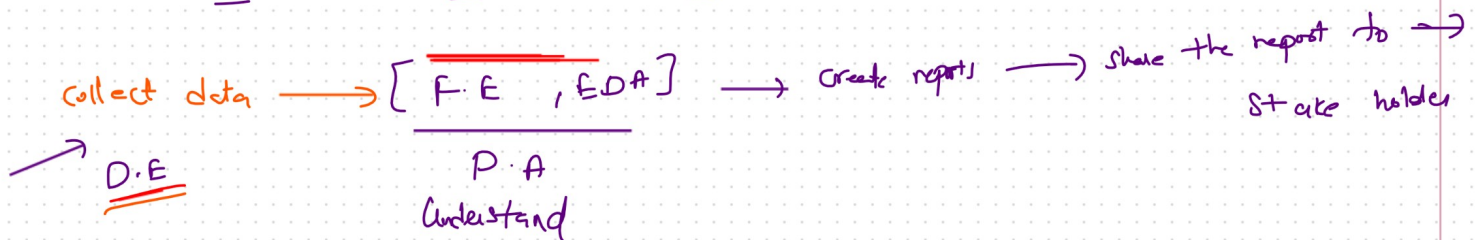
∴ different dataset → distributions (we need to know these sets of distributions)

Types

- ① Bernouli Distributions
- ② Binomial Distributions
- ③ Poisson Distributions
- ④ Uniform Distributions

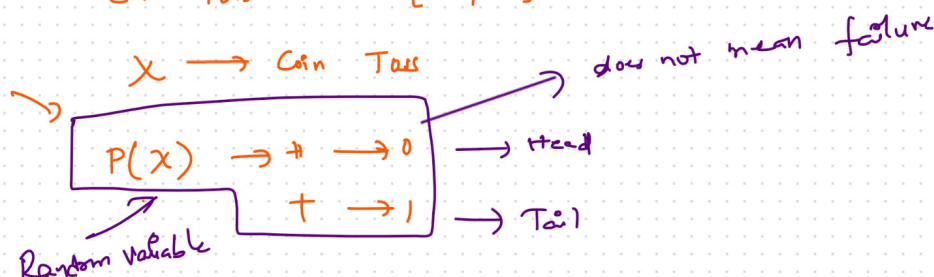
why? to understand the dataset much better
 { draw some assumptions / inferences

Dataset:	House	price	prediction	dataset		
	X_1	X_2	X_3	X_4	X_5	$\begin{bmatrix} 0 & 1 \\ T & H \end{bmatrix}$
	Size of House	No of rooms	location	Floor	Beachside	Price
	continuous random variable	discrete	discrete	discrete	0 & 1	Continuous
	pmf	pmf	pmf	pmf	pmf	pdf



① Bernoulli Distribution

Coin Toss → {H, T}



Binary $\rightarrow$ Bernoulli model

$\downarrow$
1 trail $\rightarrow$ output : 0 failure/no
1 success/yes

$\rightarrow$ Simple possible distribution

$\rightarrow$ A single trial with exactly two outcomes

0 — failure

1 — success

(1) Every coin flip $\rightarrow$ yes/no

(2) every question $\rightarrow$ true/false
pass/fail

$\rightarrow$ It has only one parameter $\rightarrow p$ (the prob of success)

example

you are applying for a job

$X \rightarrow 1 \rightarrow$ got the job prob(p)

$X \rightarrow 0 \rightarrow$ not got job prob($1-p$)

Probability formula:

$$P(X=n) = p^n (1-p)^{1-n}$$

$$X \in [0, 1]$$

$$\underline{n=0} \rightarrow p^0 (1-p)^{1-0} \rightarrow 1 \cdot (1-p) = 1-p$$

$$\underline{n=1} \rightarrow p^1 (1-p)^{1-1} = p \cdot 1 = p$$

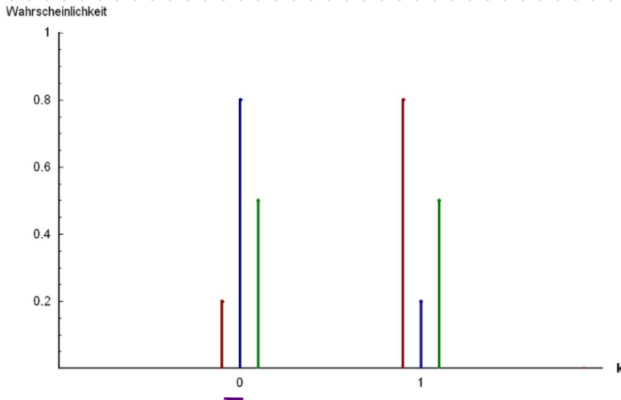
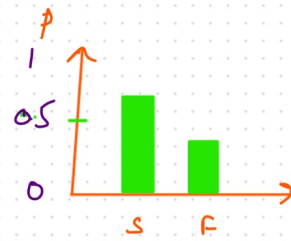
P.S

60% of customers buy shoes $\rightarrow$ on an average 60% buy shoes
 $\downarrow$
mean = 0.6

$$X \rightarrow 1 \rightarrow P(X=1) = 0.6$$

$$p \rightarrow 0.6$$

$$\begin{aligned}
 P(X=0) &= 1 - p \\
 &= 1 - 0.6 \\
 &= 0.4 \text{ (doesn't buy stock)}
 \end{aligned}$$



three examples of bernoulli distribution

Parameters

$$0 \leq p \leq 1$$

mt $q = 1 - p$

$$k = \{0, 1\} \rightarrow 2 \text{ outcomes}$$

$$Pr(\text{Success}) \Rightarrow k=1$$

$$Pr(\text{fail}) \Rightarrow k=0$$

ex Company has launched a new smart phone 'A'

$$k=1 \text{ (1) use} = 60\% \Rightarrow p$$

$$k=0 \text{ (3) not use} \rightarrow 40\% \rightarrow q = 1 - p$$

Mean of Bernoulli distribution

$$\begin{aligned}
 E(k) &= \sum_{k=0}^1 k \cdot p(k) \quad k = \{0, 1\} \\
 &= 0 \cdot 0.40 + 1 \cdot 0.60 \\
 &= 0 + 0.60
 \end{aligned}$$

$$p = 0.60$$

$$q = 0.40$$

$$\boxed{\text{mean} = 0.60} \leftarrow p$$

Median of Bernoulli distribution

median simplified

$$\begin{cases}
 \text{median} = 0 & \text{if } q > p \\
 \text{median} = 0.5 & \text{if } p = q \\
 \text{median} = 1 & \text{if } q < p
 \end{cases}$$

median

$$\begin{cases}
 0 & \text{if } p < \frac{1}{2} \\
 [0, 1] & \text{if } p = \frac{1}{2} \\
 1 & \text{if } p > \frac{1}{2}
 \end{cases}$$

$$[0, 1]$$

$$k=0, 1 \rightarrow 0.5\% \text{ } 0.5\%$$

Mode

$p > q \rightarrow p$ will be mode
else q will be mode



Variance & Standard deviation

$$k=0 \text{ \& } 1$$

$$pr(k=0) \rightarrow 0.4 \rightarrow q$$

$$pr(k=1) \rightarrow 0.6 \rightarrow p$$

$$\sigma^2 = pq$$

$$\sigma = \sqrt{pq}$$

$$\sigma^2 = 0.40 \times (0 - 0.6)^2 + 0.60(1 - 0.6)^2$$

$$\sigma^2 = 0.40 \times 0.36 + 0.60(0.16)$$

$$\sigma^2 = 0.24 \rightarrow pr(k=0) \neq pr(k=1)$$

$$q \neq p$$

$$\left. \begin{array}{l} \sigma^2 = pq \\ \sigma = \sqrt{pq} \end{array} \right\}$$

Binomial Distribution

Bernoulli is discrete with only two values

$\therefore$ model for n Bernoulli trials

Coin Toss (X-head)

H T
1 0 $\rightarrow$ coin toss

Coin Toss $p \rightarrow$ Head $\rightarrow 0.7$

$\rightarrow$ $\begin{array}{cc} 1 & 0 \\ 0 & 1 \end{array}$

$\downarrow$
N coin toss

Toss $\rightarrow$ $\begin{array}{cc} H & T \\ 1 & 0 \end{array}$

Toss $\rightarrow$ $\begin{array}{cc} 1 & 0 \end{array}$

Toss $\rightarrow$ $\begin{array}{cc} 0 & 1 \end{array}$

n -trials $\rightarrow$ n -experiments

Ques :

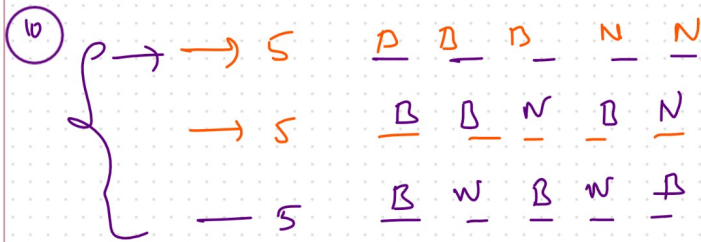
Pizza brand $\rightarrow$ garlic dip

fact

$\rightarrow$ each customer buy garlic dip along pizza with a prob of 0.6

→ Today 5 customers

→ what is the prob that exactly 3 customers will buy



→ Combinations x PMF (Bernoulli)

→ Combinations x PMF (Bernoulli)

$$\rightarrow \binom{n}{k} \times P^k (1-P)^{n-k}$$

n choose k

k → exact no. of success wanted → 3

n → total trials → 5

p → success probability p = 0.6

1-p → failure probability → 0.4

$\binom{n}{k}$ → no. of ways to arrange success among trials

Calculation:

Step 1: combination

$$= \binom{n}{k} = \frac{n!}{(n-k)! k!} = \frac{5!}{2! 3!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 3 \times 2 \times 1} = 10$$

Bernoulli → Trial 1 → answer

Trial 2 → answer

⋮

Trial n → answer



Bernoulli(p)

Binomial $\binom{n}{p}$ ^{trial} ^{success} n=10

$$T-1 + T-2 + T-3 + \dots + T-10$$

10 such trials

10x

Success part $\rightarrow p^k$
 $= (0.6)^3 = 0.216$

failure part $\rightarrow (1-p)^{n-k}$
 $\rightarrow (0.4)^2$
 $\rightarrow 0.16$

Combine all parts

$$P(X=3) = 10 \times 0.216 \times 0.16$$

$$= 0.345$$

Assignment: probability of at least 3 buyers

$$P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$$

3 or more

Quality control

eg:

f

Inspecting 10 items in a factory where each item has a 10%.

Scenario

chance of being defective

no. of trials (n) = 10

probability of success $p = 0.1$ [defective item]

No. of success (k) = varies from 0 to 10

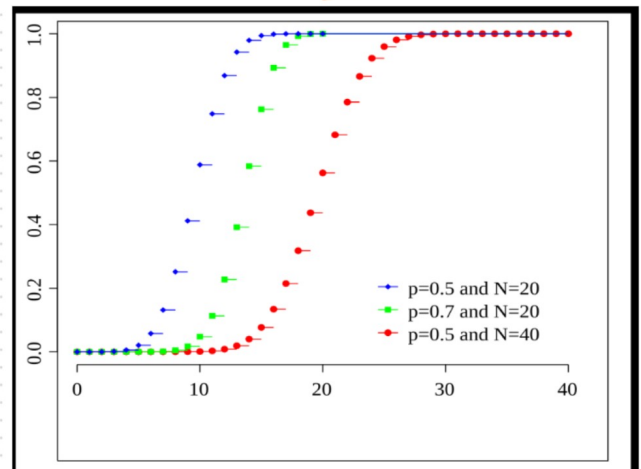
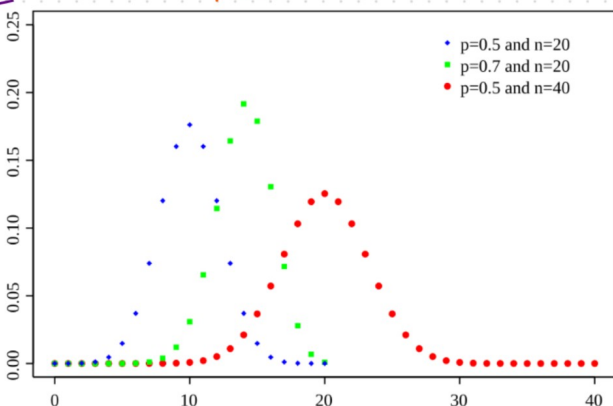
Q) what is the probability of finding exactly 2 defective items in a sample of 10?

$Pr(X=2)$?

CDF

Experiment

PMF



Notation

$B(n, p)$

parameters

$n \in \{0, 1, 2, 3, \dots, \dots\} \rightarrow$ no. of trials or experiments

$p \in [0, 1] \rightarrow$ success probability for each trial

$$q = 1 - p$$

$k \in [0, 1, 2, 3, \dots, n] \rightarrow$ no. of success
Supports

pmf

$$P_x(k, n, p) = n C_k \cdot p^k (1-p)^{n-k}$$

Mean $\rightarrow n \times p$

Variance $\rightarrow npq$

S.D $\rightarrow \sqrt{npq}$

Poisson Distribution

Model: Count of rare event occurring at a constant rate (λ) in a fixed interval

discrete: values are non-negative integers

parameters: λ (average rate of events)

memoryless: event occurs independently

web-traffic: visits per minute

Customer-service: calls per hour

} examples

formulae =
$$P(k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

$k \rightarrow$ no. of events

example: fact $\rightarrow$ A website average 4 visit per hour

$\lambda = 4$

Question what is the probability of getting exactly 6 visits in a hour

$k = 6$

$\lambda = 4$

$$P(k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!} = \frac{e^{-4} \cdot (4)^6}{6!}$$

$e = 2.7182$

$$= \frac{(2.7182)^{-4} \cdot 4^6}{6 \times 5 \times 4 \times 3 \times 2 \times 1}$$

$\approx 0.104 \quad 10\%$

Assignment:

No. of people visiting hospital every hour

fact is expected to be 3

11 Am ——— 11 pm

Q. probability of person visiting at 5 pm

$P(X=5)$

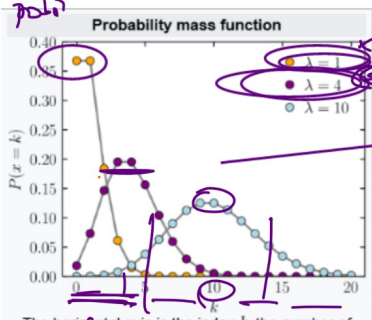
probability of person visiting between 4 pm $\rightarrow$ 5 pm

$P(X=4) + P(X=5)$

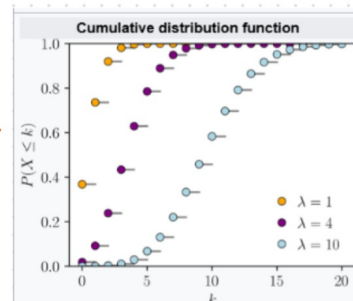
PMF

CDF

no. of user visits per hour



when λ increases bell curve



$P(X \leq k) = 1 + 2 + 3$

$$\text{Mean} = \sum(a) = \boxed{\mu = \lambda \times t}$$

$$\text{Variance} = \lambda \times t$$

$$\sigma = \sqrt{\lambda \times t}$$

λ = expected no.-of events occur at every time interval

t = time interval

Exponential distribution:

Model: the time between events in a poisson distribution



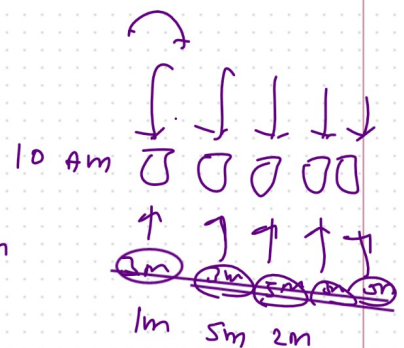
Poisson: How many events happen in a fixed time?

exponential: How much time until the next event

example

Call Center: time between 2 calls

Web server: Time between two user login



Server Request

(F) on an average, 3 request per minute API hits occur

(Q) what is the probability that the next request comes in with 10 seconds

$$\text{Formula: } P(T \leq t) = 1 - e^{-\lambda t}$$

$T \rightarrow$ waiting time until the next event
 $\lambda = \text{rate}$ $t = \text{time we care about}$

6:21

6:22:10
 t

T

$$\lambda \text{ (param rate)} = 3 \text{ req/min}$$

$$\rightarrow \text{rate per second} = \frac{3}{60} = 0.05$$

$$\lambda = 0.05$$

$$t = 10 \text{ sec}$$

$$P(T \leq 10) = 1 - e^{-0.05 \times 10}$$

$$= 1 - (2.7182)^{-0.05 \times 10}$$

$$\approx 0.393$$

$\therefore$ 39% chance of getting next request within 10 seconds

Q2 f A server crashes on an average of 0.1 times per day

Q: what is the probability it runs > 10 days without crashing

$$\lambda = 0.1$$

$$t = 10$$

$$P(T) = 1 - e^{-\lambda \cdot t}$$

$$= 1 - e^{-0.1 \times 10} \rightarrow 1 - (2.7182)^{-1}$$

$$= 1 - \frac{1}{2.7182} = 1 - 0.367$$

$$\approx 0.6321$$

$$1 - P(T) = 1 - 0.6321$$

$$= 0.367 \quad \text{not crash}$$